TASK 14 – S3 STATIC WEBSITE HOSTING

OBJECTIVE

1. Create an S3 bucket.
2. Enable static website hosting for the bucket.
3. Create and upload a simple `index.html` file.
4. Configure the required bucket permissions.
5. Verify the website using the S3 website endpoint.

PART 1 – CREATE S3 BUCKET

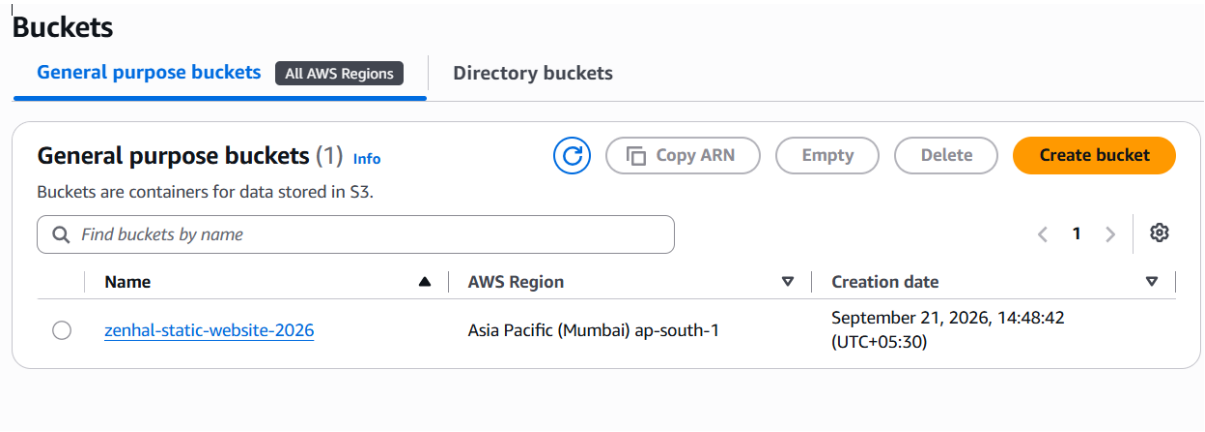
Step 1 — Create the S3 Bucket

The AWS S3 service was opened from the AWS Management Console.

A new S3 bucket was created with a unique bucket name.

The bucket was created successfully.

Note: An EC2 instance was not required for this task because S3 provides static website hosting as a managed service.



PART 2 – ENABLE STATIC WEBSITE HOSTING

Step 2 — Enable Static Website Hosting

The created S3 bucket was opened and the **Properties** tab was selected.

Under **Static website hosting**, the following configuration was selected:

```
Static website hosting: Enable
Hosting type: Host a static website
Index document: index.html
```

The configuration was saved successfully.

After enabling website hosting, AWS provided a **Bucket website endpoint**.

PART 3 – CREATE AND UPLOAD STATIC WEBSITE

Step 3 — Create `index.html`

A simple HTML file named:

`index.html`

was created with the following content:

```
<!DOCTYPE html>
<html>
<head>
    <title>Zenhal S3 Website</title>
</head>
<body>
    <h1>Welcome to My S3 Static Website</h1>
    <p>This website is hosted using Amazon S3.</p>
    <p>Cloud & DevOps Practice</p>
</body>
</html>
```

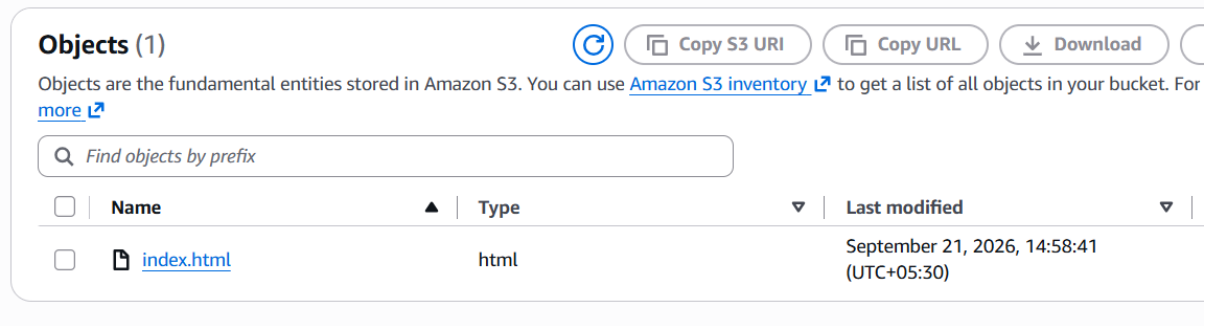
Step 4 — Upload `index.html`

The S3 bucket was opened and the **Objects** section was selected.

The `index.html` file was uploaded using:

Upload → Add files → Select `index.html` → Upload

After the upload, `index.html` was visible inside the bucket.



PART 4 – CONFIGURE BUCKET PERMISSIONS

Step 5 — Configure Public Access

The **Permissions** tab of the S3 bucket was opened.

The **Block all public access** setting was already unchecked.

This allows the bucket to be configured for public website access.

Step 6 — Add Bucket Policy

A bucket policy was added to allow visitors to read the objects from the bucket.

The policy used was:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

The bucket name in the `Resource` was replaced with the actual S3 bucket name.

The policy was successfully added.

The `s3:GetObject` permission allows users to **read the website files** from the bucket.

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects own

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::zenhal-static-website-2026/*"
    }
  ]
}
```

PART 5 – VERIFY THE WEBSITE

Step 7 — Open the Website Endpoint

The **Properties** tab was opened again and the **Static website hosting** section was checked.

The **Bucket website endpoint** was opened in a web browser.

The webpage was displayed successfully with:

```
Welcome to My S3 Static Website

This website is hosted using Amazon S3.

Cloud & DevOps Practice
```

This confirmed that the S3 static website was working correctly.



Welcome to My S3 Static Website

This website is hosted using Amazon S3.

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VERIFICATION

The following were successfully verified:

- S3 bucket was created.
- Static website hosting was enabled.
- `index.html` was configured as the index document.
- `index.html` was uploaded to the bucket.
- Required public access was configured.
- Bucket policy allowed `s3:GetObject`.
- S3 website endpoint successfully displayed the webpage.

CONCLUSION

The S3 static website hosting task was successfully completed using the AWS Management Console. A simple `index.html` webpage was hosted in an S3 bucket, the required permissions were configured, and the website was successfully verified using the S3 website endpoint.